A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 34

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 2 points below. **1** Landfill fell from 79% to 14% (large decrease) **1** Incineration rose from 9% to 42% and recycling / composting rose from 11% to 43% (now the largest) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Incineration** → burning waste, usually with energy recovery • **recycling** → reprocessing a material to make something new **Do not credit** • 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	2

Do not write outside the box

Question	Marking guidance	Marks
01.2	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** Organic matter decays anaerobically releasing methane (and CO2) - greenhouse gases **1** Toxic leachate may leak into groundwater / rivers from poorly managed sites **1** Uses large areas of land – loss of habitats / farmland – contamination prevents later use **1** Resource value of metals, glass, plastics is lost **1** Transport noise / congestion / litter / pests / odour **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **leachate** → polluted liquid draining out of waste **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	3
01.3	**Any 4 of 7** 1 mark each for any 4 of the 7 points below, in any order, maximum 4. **1** Polymer / clay liner to prevent escape of leachate **1** Collection and treatment of leachate **1** Collection of methane and use as a fuel **1** Impermeable cap (clay and polymer) when complete to stop water entering **1** Regular covering with soil (pests, odour) – deodorising sprays **1** Perimeter fence to trap wind-blown litter **1** Separation and recording of waste types – reactive / flammable wastes not dumped together **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **leachate** → polluted liquid draining out of waste • **Impermeable** → does not let water through **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	4
01.4	1 mark for the point below. **1** Mixed / contaminated / dirty wastes (plastics, dirty paper) are difficult and costly to separate and recycle but can be burnt with energy recovery **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **incineration** → burning waste, usually with energy recovery • **recycling** → reprocessing a material to make something new **Do not credit** • 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new	1

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 280-282; AQA spec 3.4.3.2.12.	
02.1	**1 mark each** 1 mark for each of the 2 points below. **1** Lowest overnight (about 25 GW at 04:00), rising steeply from 06:00 to about 40 GW by 08:00 **1** Fairly steady through the day then a peak of about 50 GW at 18:00 (evening) before falling overnight **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
02.2	**Marks shared** Up to 4 marks in total - share them between the 3 points below according to how completely each one is made. **up to 2** 24-hour day / night cycle: people asleep overnight – morning rise as heating, lighting, cooking, industry and offices start **up to 2 Evening peak:** lighting, cooking, heating as people return home – short-term 'TV pickup' at programme breaks / mealtimes **up to 2 Other causes:** seasonal (winter heating / lighting, summer air conditioning) – weather-related (cold snaps) – weekday / weekend differences in industrial use **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** They generate 24 hours a day because it is uneconomic / slow to turn them off and on, and output changes slowly **1** Demand falls faster at night than output can be reduced, so the surplus must be stored or is lost as heat from cables **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do	2

Do not write outside the box

Question	Marking guidance	Marks
02.4	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1 Intermittent renewable resources:** wind, solar, tidal **1** Bulk delivery of transported fuels (oil tankers, coal trains, biofuel harvests) **1** Breakdowns / maintenance of power stations **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Intermittent** → only generates when the wind blows or the Sun shines / supply is not continuous **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 221-222; AQA spec 3.3.4.1.	
03.1	**1 mark each** 1 mark for each of the 6 points below. **1 Location:** discharge into the sea / large water bodies where currents dilute and disperse the pollutant **1 Location:** chimneys / emissions sited downwind of urban areas so pollutants are not carried over populations – avoiding valleys and enclosed water bodies where pollutants concentrate **1 Location:** not discharging wastes onto permeable rock above an aquifer / not siting landfill on aquifers **1 Timing:** restrict polluting activities (industrial combustion, vehicle use) during temperature inversions when dispersal is poor **1 Timing:** discharge liquid effluent when river flow is high / on an ebb tide so it is diluted and carried away **1 Timing:** pesticide spraying at night / on still days – noisy operations restricted to daytime – night-flight restrictions **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **permeable** → how easily water passes through / lets water through • **aquifer** → a permeable rock that stores groundwater / underground water store **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	6

Do not write outside the box

Question	Marking guidance	Marks
03.2	**1 mark each** 1 mark for each of the 2 points below. **1** The organisation / person that causes the pollution is responsible for (paying for) the damage caused **1** If the cost of preventing the pollution is less than the cost of the damage / fines, there is a financial incentive to prevent it – works best when penalties are high and escape is unlikely **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** It is assumed that the waste will cause pollution / harm until research shows it is unlikely to **1** So release is not permitted until it has been tested – lack of knowledge of a problem is not an excuse / does not reduce responsibility **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • the precautionary principle described as banning anything new - it shifts the burden of proof onto showing that harm is unlikely	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-245; AQA spec 3.4.3.1.	
04.1	**1 mark each** 1 mark for each of the 2 points below. **1** The lowest ore purity that can be mined economically with existing technology **1** Deposit X (0.9% is above 0.4%; Y at 0.35% is part of the resource, not the reserves) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
04.2	**1 mark each** 1 mark for each of the 6 points below. **1** X has higher grade so less rock mined, less spoil, less energy and pollution per tonne of copper **1 But X is deep:** costs rise more than proportionally with depth (sides cannot be vertical so much more rock removed; more water inflow and pumping) **1** X has hard overburden requiring blasting (cost, noise, dust) and high rainfall increasing drainage costs **1** X is far from transport infrastructure so transport costs are high **1** Y is shallow with loose overburden (cheap to remove but landslide risk so gentler slopes / larger land take), near a railway, low rainfall **1** Y's grade is below the COOG so it is not viable now but would become viable if copper prices rose or extraction technology improved (eg bioleaching) – both sulfides risk acid mine drainage **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	6
04.3	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Higher market value / price of the metal **1** Improved extraction technology **1** Lower energy costs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 138-139; AQA spec 3.2.3.5.	
05.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** (60 x 45) / 9 **1** = 300 **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (beetles) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2

Do not write outside the box

Question	Marking guidance	Marks
05.2	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Marked individuals mix freely / randomly with the rest of the population **1** No births, deaths, immigration or emigration between samples **1** Marking does not affect survival or behaviour (eg make them more visible to predators) **1** Marks do not wear off / fall off **1** All individuals have the same chance of being caught (traps do not attract or repel marked animals) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Lincoln Index** → estimating population size from the proportion of marked animals recaptured **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • 'pick places at random by eye' - positions have to be generated with random numbers to remove bias	4
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** Fewer marked beetles would survive to be recaptured **1** So the proportion marked in the second sample would be too low and the population would be overestimated **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
05.4	**1 mark each** 1 mark for each of the 2 points below. **1** Enough time for marked beetles to mix with the population **1** Not so long that births / deaths / migration or loss of marks change the population **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 401-402; AQA spec 3.7.2.1.	

Do not write outside the box

Question	Marking guidance	Marks
06.1	**1 mark each** 1 mark for each of the 5 points below. **1 Double hull:** a gap of up to 2 m between the hulls so damage to the outer hull in a collision / grounding does not release oil **1 Inert gas system:** cooled engine exhaust gas (no oxygen) fills the space above the oil so an explosive oil vapour / air mixture cannot form **1 Separate ballast tanks:** ballast water is not carried in oil tanks so no residual oil is discharged when ballast is pumped out **1 Twin engines / rudders:** if one fails the ship can still be steered / propelled and is not carried onto rocks by currents or wind **1 Crude oil washing:** sludges are mixed and removed with the cargo so tanks do not need washing with seawater that was then discharged – oily water is unloaded at the terminal **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	5
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** Offshore routes keep tankers further from the coast so if there is engine failure there is more time to deal with it before the ship is carried ashore **1** GPS gives accurate navigation – AIS lets ship positions be monitored and coordinated to avoid collisions and groundings **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **accurate** → close to the true value	2
06.3	**1 mark each** 1 mark for each of the 2 points below. **1** The tank stands on an impermeable base surrounded by a wall **1** The volume enclosed by the wall is greater than the tank volume so all the oil is contained if the tank splits / cannot overflow **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **impermeable** → does not let water through	2
06.4	1 mark for the point below. **1** Separates floating oil from drainage water so the oil is retained (for later treatment) and only water flows to the drains / river **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 259-260; AQA spec 3.4.3.2.7.	

Do not write outside the box

Question	Marking guidance	Marks
07.1	**1 mark each** 1 mark for each of the 4 points below. **1 Potable:** no pathogens, toxins at acceptable levels, looks / smells / tastes good - public health, consumer complaints **1 Spray irrigation:** low turbidity, low toxins such as heavy metals - blockage of pipes, contamination of food **1 Condenser cooling:** no gross solids - pipe blockage **1 Boiler water:** no dissolved minerals / soft water - mineral deposits reduce heat exchange and block pipes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	4
07.2	**1 mark each** 1 mark for each of the 2 points below. **1** Raises dissolved oxygen and removes hydrogen sulfide (bad-egg smell) from decay of organic matter **1** Some dissolved toxic / bad-tasting metals become insoluble when oxidised and can be removed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
07.3	**1 mark each** 1 mark for each of the 2 points below. **1** Clay particles carry electrostatic charges that make them repel each other so they stay suspended **1** Flocculants such as aluminium sulfate (alum) or polyelectrolytes neutralise the charges so the particles clump and settle in the clarifier **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
07.4	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Chlorine **1** Ozone **1** Ultraviolet light **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 126-127; AQA spec 3.2.2.4.	
08.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B - tiny pores between clay particles allow capillary rise – none in sand	1

Do not write outside the box

Question	Marking guidance	Marks
08.2	**1 mark each** 1 mark for each of the 3 points below. **1 Hygroscopic:** held tightly by adhesive forces on particle surfaces – cannot drain and cannot be absorbed by plants **1 Capillary:** held in pores by surface tension – does not drain but can be absorbed by plants **1 Gravitational:** drains freely through large pores – available to plants only briefly as it passes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
08.3	1 mark for the point below. **1** Small round crumbs leave spaces giving good drainage, aeration and easy root penetration – large flat platy peds pack together reducing all three **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **peds** → soil crumbs / aggregates of soil particles	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-169; AQA spec 3.2.5.	
09.1	**1 mark each** 1 mark for each of the 4 points below. **1 Wheel vibration:** track polishing / grinding, sound-absorbing ballast, sound-absorbing suspension **1 Pantograph:** aerodynamic fairing **1 Wheel squeal:** lubrication of wheels / track on corners **1 Braking:** composite-material brakes instead of cast iron **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
09.2	1 mark for the point below. **1** Roads pass through / near almost all residential areas and traffic is continuous through the day, unlike airports or railways which affect limited areas **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1

Do not write outside the box

Question	Marking guidance	Marks
09.3	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** Sound-absorbing road surfaces (porous asphalt / rubber-crumb asphalt) **1** Acoustic fences, embankments / baffle mounds, walls or dense tree planting **1** Traffic management for free flow (less acceleration noise) – re-routing traffic away from residential areas **1** Double glazing / acoustic insulation of buildings **1 Vehicle design:** better aerodynamics, engine insulation, quieter exhausts, quieter tyres **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
09.4	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Air bubble curtains around the pile driver absorb noise **1** Acoustic survey to detect cetaceans first and delay work if present – increase volume gradually (soft start) so animals can leave **1** Drilling instead of pile-driving **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'wind turbines can generate all the time' - output depends on wind speed, and they shut down in very high winds	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 292-294; AQA spec 3.4.3.2.13.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Fossil fuel technologies:** secondary / tertiary recovery raising recovery from 20% to 60% – directional drilling and subsea wells – ROVs – fracking for tight oil and gas – tar sands / oil shales – coal gasification / liquefaction – enhanced gas recovery – methane hydrates – CCS (pre / post combustion, storage) • **Limits:** still finite (discoveries declining since the 1980s) – rising energy and financial cost of unconventional sources – environmental impacts (aquifers, water use, tremors, land take, habitat damage) – CO2 and other pollution – CCS unproven at scale and impractical for vehicles • **Nuclear:** new uranium extraction (polymer adsorption from seawater, phosphate mining, coal ash) – molten salt reactors – plutonium breeder and thorium reactors increasing fuel use many-fold – fusion as a long-term prospect • **Limits:** cost overruns, decommissioning, waste, public opposition, proliferation, embodied energy of low-grade uranium – fusion not commercial • **Sustainability criteria:** maintaining supply without unacceptable environmental, economic or social impact – comparison with renewables and energy conservation – energy gap argument • **Judgement:** technologies can extend supplies and reduce some impacts but cannot make non-renewables sustainable indefinitely	9

Do not write outside the box

Question	Marking guidance	Marks
	– role as transition / base-load alongside renewables **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **fracking** → pumping fluid into rock at high pressure to release gas or oil • **CCS** → capturing carbon dioxide from flue gases and storing it underground • **adsorption** → nutrient ions sticking to the surfaces of clay and humus **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	
	Model answer - Level 3 (8 marks)	
	Technology has repeatedly extended fossil-fuel supplies. Secondary and tertiary recovery raise the proportion of oil recovered from about 20% to 60%; directional drilling, subsea wells and ROVs open deep-water fields; fracking releases tight oil and gas; tar sands and oil shales, coal gasification and liquefaction convert lower-grade resources; and carbon capture and storage could reduce the CO_2 impact. Nuclear fission can be stretched even further: uranium can be adsorbed from seawater with polymers or recovered from phosphate mining and coal ash, molten-salt reactors run more safely, and plutonium breeder and thorium reactors multiply the energy obtainable from each tonne of fuel, with fusion a long-term prospect.	
	None of this makes the resources renewable. Discoveries of conventional oil have declined since the 1980s, unconventional sources cost more energy and money per unit and damage aquifers, use huge volumes of water, cause tremors and take land, and CCS is unproven at scale. Nuclear projects overrun their budgets, decommissioning and long-lived waste remain unsolved, public opposition and proliferation constrain expansion, low-grade uranium has a high embodied energy, and fusion is not commercial.	
	Sustainability means maintaining supply without unacceptable environmental, economic or social impact. New technology can extend supplies for decades and cut some impacts, which matters for closing the energy gap while renewables and conservation scale up, but it cannot make non-renewable resources sustainable indefinitely. Their proper role is as a transition and base-load source alongside renewables, not a permanent solution.	
	Weaker answer - Level 2 (5 marks)	
	New technologies mean fossil fuels can last longer. Fracking gets gas out of shale rock and directional drilling reaches oil under the sea. Carbon capture and storage can stop the carbon dioxide going into the atmosphere. For nuclear power, breeder reactors and thorium reactors can use fuel more efficiently and fusion might work in the future. However fossil fuels are still non-renewable so they will run out eventually, and fracking can pollute groundwater. Nuclear waste is still a problem. So new technologies help but they cannot make fossil fuels and nuclear power fully sustainable.	
	What makes the difference	

Do not write outside the box

Question	Marking guidance	Marks
	• Level 2 reaches a sensible conclusion but supports it thinly; Level 3 gives specific evidence (20% to 60% recovery, declining discoveries, embodied energy of low-grade uranium).	
	• Level 3 sets out a criterion for sustainability and judges against it - the hallmark of 'evaluate'.	
	• Level 3 places the technologies in context (transition and base load alongside renewables) rather than treating them in isolation.	
10.2	1 mark for the point below. **1** 80-90% **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 183-196; AQA spec 3.3.3-3.3.4.1.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Discuss the difficulties in monitoring and predicting global climate change and evaluate the methods used to overcome them.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Nature of the problem:** interconnected atmosphere-hydrosphere-biosphere systems – changes over different time and spatial scales – natural fluctuations (solar output, orbit) masking or exaggerating anthropogenic change – time delay between cause and effect (ocean heat capacity) • **Limitations of historic data:** sparse global coverage, unsophisticated equipment, heat island bias, no atmospheric composition records • **Proxy data and their reliability:** dendrochronology, coral growth rings, pollen in dated sediments, ice cores (gas bubbles, isotope ratios, volcanic ash layers, compaction problems) • **Modern monitoring:** long-term data sets (Mauna Loa), electronic equipment, satellites (LEO vs geostationary), Argo floats, remote sensing of ice thickness / area / sea level • Feedback mechanisms and tipping points adding uncertainty – ecological uncertainty (species survival, distribution, fragmentation) • **Computer models:** how they are tested and refined – importance of accurate representative data – scenario-based predictions and ranges	25
	• Uncertainty over ENSO / Conveyor / jet stream responses • **Evaluation and conclusion:** confidence in the overall trend and human cause despite detailed uncertainty – role of peer review and international bodies (IPCC) *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 89-90, 100-104, 10; AQA spec 3.2.1.2, 3.1.1.3.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Discuss how knowledge of the properties of energy resources can be used to reduce the severity of their impact on the environment.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Properties:** energy density, renewability, abundance, location, intermittency / predictability, storage, transport, conversions, technology level, environmental impacts • **Fossil fuels:** choosing lower-impact fuels (gas over coal), desulfurisation, CCS, efficient combustion, siting power stations near cooling water and coal ports – new extraction technologies (directional drilling, secondary / tertiary recovery) and their impacts (fracking, tar sands) • **Nuclear:** high energy density reduces transport – waste properties (half-life) determine storage – thorium and fusion prospects • **Renewables:** matching resource to location (wind, HEP, solar, wave, tidal, geothermal, biofuel siting) – intermittency managed by storage (pumped HEP, batteries, hydrogen, P2G, V2G), grid management and diversity – low-impact designs (helical turbines, VAWT, tidal lagoons) • **Energy conservation reduces demand:** transport, buildings, industry (CHP, heat exchangers) • Impacts before and during use	25
	– life-cycle / embodied energy • **Evaluation:** no resource is impact-free – knowledge allows the best mix and mitigation – judgement *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-235; AQA spec 3.3.2-3.3.4.	

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